Appendix

A1: Social optimum with state-centred perspective

Maximising the sum of the valuations of both countries:

$$G^* = V_1 + V_2$$
$$= \alpha_1 r_1 - \psi_1 r_1^2 + \lambda_1 r_2 - p_1 r_1 + \alpha_2 r_2 - \psi_2 r_2^2 + \lambda_2 r_1 - p_2 r_2$$
$$= \alpha_1 r_1 - \psi_1 r_1^2 + \lambda_1 (N - r_1) - p_1 r_1 + \alpha_2 (N - r_1) - \psi_2 (N - r_1)^2 + \lambda_2 r_1 - p_2 (N - r_1)$$

From the first order conditions of G^*, we can derive the share of refugees each country should accept

$$\frac{\partial G^*}{\partial r_1} = 0 \rightarrow r_1^* = \frac{(\alpha_1 - \alpha_2) + (\lambda_2 - \lambda_1) + \ (p_2 - p_1) + 2\psi_2 N}{2(\psi_1 + \psi_2)}$$

and

$$r_2^* = N - \ r_1^* = \frac{(\alpha_2 - \alpha_1) + (\lambda_1 - \lambda_2) + \ (p_1 - p_2) + 2\psi_1 N}{2(\psi_1 + \psi_2)}$$

If the parameters are symmetric for both countries ($\alpha_1 = \alpha_2, \lambda_1 = \lambda_2$ and $\psi_1 = \psi_2$), the overall costs would be lowest if the number of refugees were split evenly with $r_1^* = \frac{2\psi N}{2(2\psi)} = \frac{N}{2} = \tilde{r}_{1,2}$. On the other hand, $r_1^* \rightarrow 0$, if $(\alpha_1 - \alpha_2) + (\lambda_2 - \lambda_1) + \ (p_2 - p_1) + 2\psi_2 N = 0$. Rearranging the formula yields $(\alpha_2 - \alpha_1) - (\lambda_2 - \lambda_1) + \ (p_1 - p_2) = \ 2\psi_2 N$. The right-hand side has to be > 0, because $\psi_2$and N are positive. For the left-hand side, we know that $(\alpha_2 - \alpha_1) - (\lambda_2 - \lambda_1) < 0$, if $(\alpha_2 - \alpha_1) + \lambda_2 > \lambda_1$. In that case $(p_1 - p_2)$ has to be >> 0 (and vice versa otherwise). Thus, $r_1^* \rightarrow 0$ if the linear costs for the reception are higher for 1 than for 2 and the joint utility of 1 and 2 for receiving the refugees in 2 is higher.

A2: Public good characteristics

For country 1 it is optimal to aim for ${r_1}' \rightarrow$ 0 and ${r_2}' \rightarrow$ N. This is observable if we compare $V_1(r_1^*)$ with $V_1(0) = \lambda_1 N$. We have:

$$\lambda_1 N \leq \alpha_1 r_1 + \lambda_1 (N - r_1) - \psi_1 {r_1}^2 - p_1 r_1$$

and thus:

$$0 \leq \alpha_1 - \lambda_1 - \psi_1 r_1 - p_1,$$

This is a contradiction, since we assume $\alpha_i < \lambda_i$. Thus, free-riding is the dominant behaviour for both countries and depicts a typical prisoner's dilemma situation.

A3: Social optimum, including refugees' preferences

Maximising the sum of the valuations of all actors:

$$\begin{gathered} G^{**} = V_1 + V_2 + V_N \\ = \alpha_1 r_1 - \psi_1 r_1^2 + \lambda_1 (N - r_1) - p_1 r_1 \\ + \alpha_2 (N - r_1) - \psi_2 (N - r_1)^2 + \lambda_2 r_1 - p_2 (N - r_1) + \sum_{i=1}^{N} V_N^i \end{gathered}$$

This leads to the following distribution of refugees among countries:

$$\frac{\partial G^{**}}{\partial r_1} = 0 \rightarrow r_1^{**} = \frac{(\alpha_1 - \alpha_2) + (\lambda_2 - \lambda_1) + \ (p_2 - p_1) + 2\psi_2 N + \dfrac{\partial V_N}{\partial r_1}}{2(\psi_1 + \psi_2)}$$

and

$$r_2^{**} = N - \ r_1^{**} = \frac{(\alpha_2 - \alpha_1) + (\lambda_1 - \lambda_2) + \ (p_1 - p_2) + 2\psi_1 N + \dfrac{\partial V_N}{\partial r_2}}{2(\psi_1 + \psi_2)}$$

A4: Public good characteristics (three players)

We compare $V_1(r_1^{**})$ with $V_1(0) = \lambda_1 N$. $\lambda_1 N \leq \alpha_1 r_1^{**} + \lambda_1 (N - r_1^{**}) - \psi_1 {r_1^{**}}^2 - p_1 r_1^{**}$ if $0 \leq \alpha_1 - \lambda_1 - \psi_1 r_1^{**} - p_1 r_1^{**}$, which is a contradiction, since we assume $\alpha_i < \lambda_i$.